UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Course Code : SAIA 2113

Course Name : Machine Learning

Type of Assessment : Assignment – Supervised learning and Unsupervised Learning (pair work)

Full Marks : 100 Marks (30%)

Date : 5 December 2025

Submission deadline : Before 12.00 a.m. on Friday, 5 December 2025

This assignment consists of five parts. Answer all questions.

Submit the assignment via E-Learning.

Include the coding file in. ipynb format.

Name	
Matric ID	
Section	
Lecturer's Name	

Part 1: Dataset (15 marks)

Task: Create an image dataset for classification problems.

Requirements:

1. At least 2 classes (e.g., Positive or negative, normal or abnormal, yes or no, 0 or 1, Cats vs Dogs).

(5 marks)

2. Minimum 50 images per class. (If you have 2 classes, then the number of images should be 100).

(5 marks)

3. You can make your own image collection by taking pictures with your phone or by using free images from websites like Unsplash or Wikimedia. Feel free to mix both your own photos and ones you find online.

(5 marks)

Part 2: Supervised Learning (25 marks)

Task: Apply one supervised learning technique.

Requirements:

1. Preprocess by resizing and normalizing the images. Write the code and discuss the process.
(5 marks)

2. Split into training and testing sets. Write the code and discuss the process.

(5 marks)

3. Use one supervised learning technique. Write the code and discuss the process.

(5 marks)

4. Train the model to classify your images. Write the code and discuss the process.

(5 marks)

5. Evaluate with performance metrics such as accuracy, precision, recall, and F1-score. Write the code and discuss the process.

(5 marks)

Part 3: Unsupervised Learning (25 marks)

Task: Apply k-means clustering technique.

Requirements:

1. Apply k-means clustering technique. Write the code and discuss the process.
   (20 marks)

2. Compare the clusters generated by k-means clustering with only one actual image for each class. Discuss.
   (5 marks)

Part 4: Reflection & Comparative Analysis (25 marks)

Task: Analyse and discuss your findings.

Requirements:

1. Based on the task given on part 2 and 3, what are the differences between the supervised learning technique that you applied with the k-means clustering technique?
   (**5 marks)**

2. What were the biggest challenges in data collection and data preprocessing?
   (5 marks)

3. How well does supervised learning perform? Discuss.
   (5 marks)

4. Discuss the output of the clusters generated by k-means with the actual image?
   (5 marks)

5. What would you improve if you had more time or data?

   (5 marks)

Part 5: Report requirement (10 marks)

Task: Prepare a report with Turnitin and AI checker percentage less than 15%.

Requirements:

1. Clear and concise explanations.
   (2 marks)
2. Well-organized presentation of results.
   (2 marks)
3. Citations for any external sources.
   (2 marks)
4. Maximum 10 pages including figures (proper figure labels and captions) and code snippets.
   (2 marks)
5. Include all plots and images referenced in your analysis.
   (2 marks)

Grading rubric – Part 1: Dataset

Criteria	1 – Poor	2 – Fair	3 – Good	4 – Very Good	5 – Excellent
Q1 – Number of Classes (At least 2 classes)	No class.	One class only.	Two classes only and unclear label.	Two classes and clearly labelled.	More than two well-defined classes and clearly labelled.
Q2 – Number of Images per Class (Min. 50 per class)	Far fewer than 19 images per class.	20–29 images per class.	30–39 images per class.	40–49 images per class.	Meets minimum (50 per class or more).
Q3 – Image Source Variety (Own photos and/or copyright-free images)	No effort to collect own images or using copyright images.	Have the effort to collect own images or using copyright images, but limited variation.	Some variations of own images and copyright-free sources.	Variety of own images and copyright-free sources.	Excellent variety of own images and copyright-free sources.

Grading rubric – Part 2: Supervised Learning

Criteria	1 – Poor	2 – Fair	3 – Good	4 – Very Good	5 – Excellent
Q1 - Image Preprocessing (Resize & Normalize)	No preprocessing done.	Either resizing or normalizing only have been applied.	Resizing and normalization applied, but other parts may be missing, unclear, or partially working.	Images properly resized and normalized with minor issues. E.g., missing comments or slight inconsistencies.	All images consistently and correctly pre-processed (resized + normalized).
Q2 - Data Splitting (Train/Test Split)	No train-test split applied.	Train/test split applied, but no explanation.	Train/test split done with explanation but no justification of ratio.	Good train/test split with justification of ratio.	Properly implemented train/test split with stratification and clear reasoning.
3. Supervised Learning Technique Used	No model applied.	Incorrect model with no parameter tuning.	Incorrect model with parameter tuning.	Correct model with parameter tuning.	Effective supervised model with well-tuned parameters.
4. Model Training (Classify Images)	Model not trained or unsuccessful run.	Basic model training attempted with minimal results (below than or equal to 60%).	Model trained with some success on image dataset. (61% to 70%).	Model trained effectively with good classification output. (71% - 89%).	Model shows strong performance (90% or above) or visuals (loss/accuracy curves).
5. Model Evaluation (Accuracy, Precision, Recall, F1)	No metrics reported.	One or two metrics used, but poorly interpreted.	Only three metrics used but poorly interpreted.	All four metrics used but weak discussions.	Excellent evaluation with metrics and thoughtful analysis.

Grading rubric – Part 3: Unsupervised Learning

Criteria	2 – Poor	4 – Fair	6 – Good	8 – Very Good	10 – Excellent
Q1 – Application of K-Means Clustering	No model applied.	Incorrect model with no parameter tuning.	Incorrect model with parameter tuning.	Correct model with parameter tuning.	Effective unsupervised model with well-tuned parameters.
Criteria	**1 – Poor**	**2 – Fair**	**3 – Good**	**4 – Very Good**	**5 – Excellent**
Q2 – Comparison with Actual Image Samples (One per class)	No comparison.	One class comparison only.	Two or more classes comparisons made, but lacks of discussion.	Two or more classes comparisons made with discussion.	Excellent comparison with insightful interpretation and clear visual evidence per class with clear discussion.

Grading rubric – Part 4: Reflection & Comparative Analysis

Criteria	1 – Poor	2 – Fair	3 – Good	4 – Very Good	5 – Excellent
Q1 – Comparison of supervised vs k-means	No clear understanding; vague or irrelevant comparison	Basic difference mentioned, lacks clarity or depth	Adequate comparison with some technical explanation	Good explanation of differences with examples from their task	Clear, insightful, and well-structured comparison showing deep understanding with some technical explanation and examples from their task.
Q2 – Challenges in data collection & preprocessing	No challenges described or very vague.	Mentions one challenge only.	Mentions two challenges only.	Mentions three challenges with limited details.	Mentions three challenges with meaningful discussion.
Q3 – Performance of supervised learning	No interpretation.	Incorrect interpretation of the performance.	Correct interpretation of the performance but very little analysis.	Correct interpretation of the performance with good discussions.	Correct interpretation of the performance with in-depth discussion with critical analysis and possible model improvement ideas.
Q4 – Interpretation of k-means clusters	No interpretation.	Incorrect interpretation of the performance.	Correct interpretation of the performance but very little analysis.	Correct interpretation of the performance with good discussions.	Correct interpretation of the performance with in-depth discussion with critical analysis and possible model improvement ideas.

Q5 – Improvements with more time/data	No suggestion or unrealistic idea	Basic suggestion with little connection to task	Relevant suggestion, somewhat justified	Well-thought improvement based on task experience	Highly relevant, practical suggestions showing strong reflection and insight

Grading rubric – Part 5: Report requirement

Criteria	1 – Fair	2 – Excellent
Q1 – Clear and concise explanations	Limited clarity with some irrelevant or vague points	Extremely clear, concise, and insightful explanations throughout
Q2 – Well-organized presentation of results	Basic organization but hard to follow	Exceptionally organized, visually appealing, and logically sequenced
Q3 – Proper citations for external sources	One source cited with unclear formatting	All external sources are clearly cited using consistent and proper format
Q4 – Document format and length (≤10 pages, proper captions)	Minor length or formatting issues, some unlabelled figures	Fully adheres to page limit, all figures are labelled with informative captions
Q5 – All referenced plots/images included	Some plots/images missing or unclear	All visuals are included, high-quality, and directly support the analysis